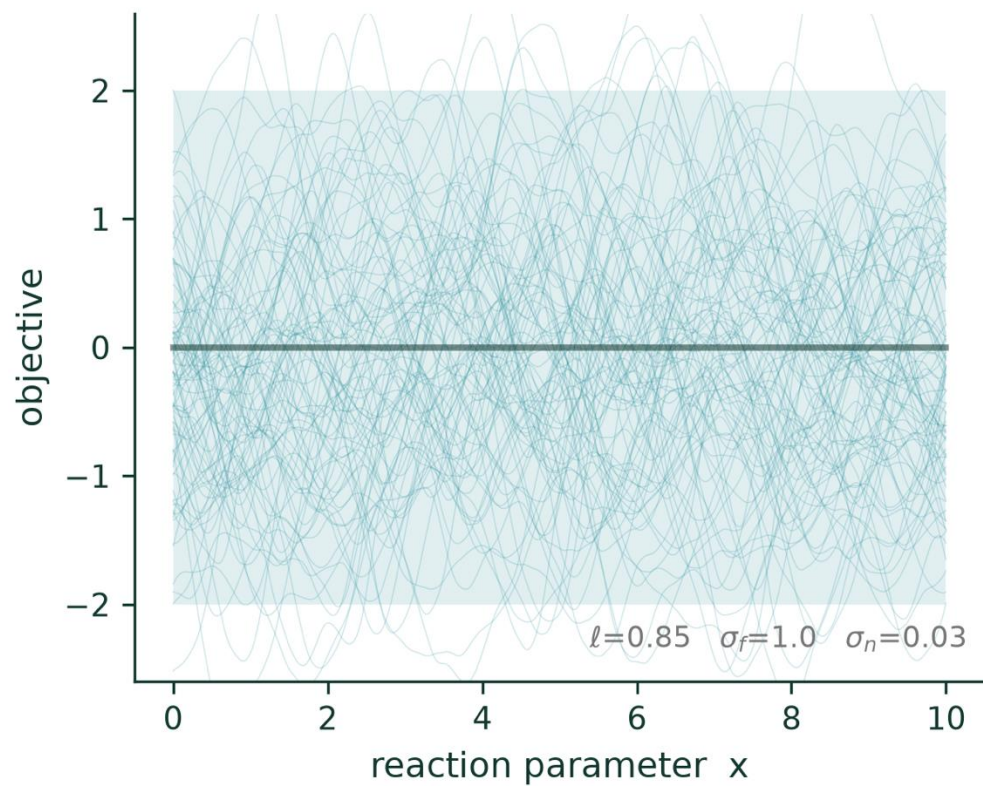
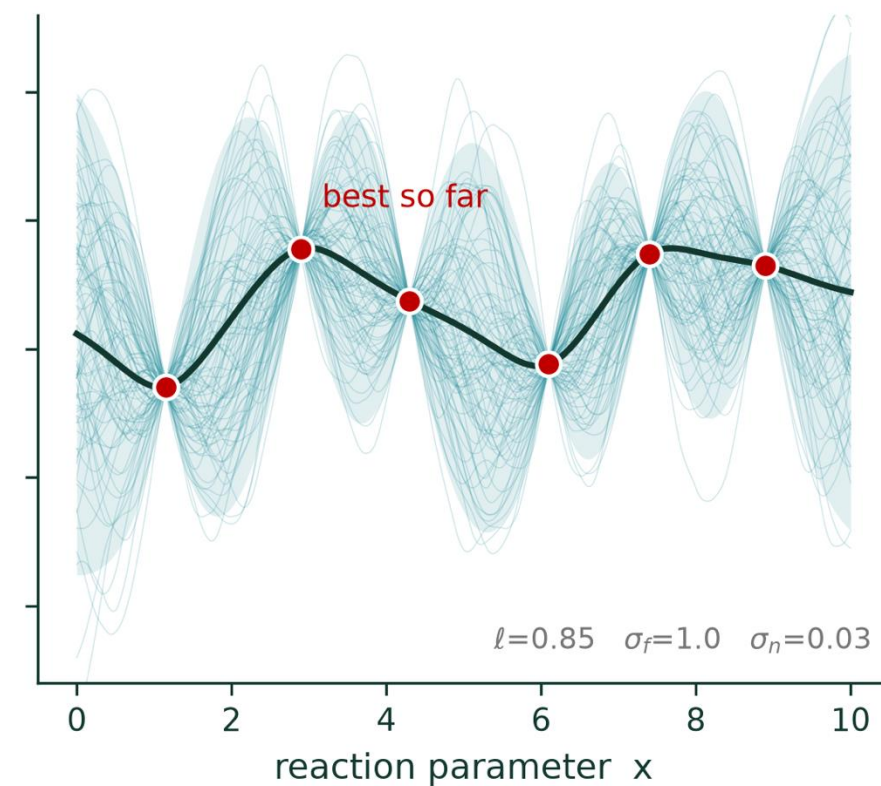


# Bayesian Optimization



DATA  
ACQUISITION



J. Schleinitz · Tuesday, August 18, 2026 · RSC 275, Caltech

Claude Code was used for slides preparation

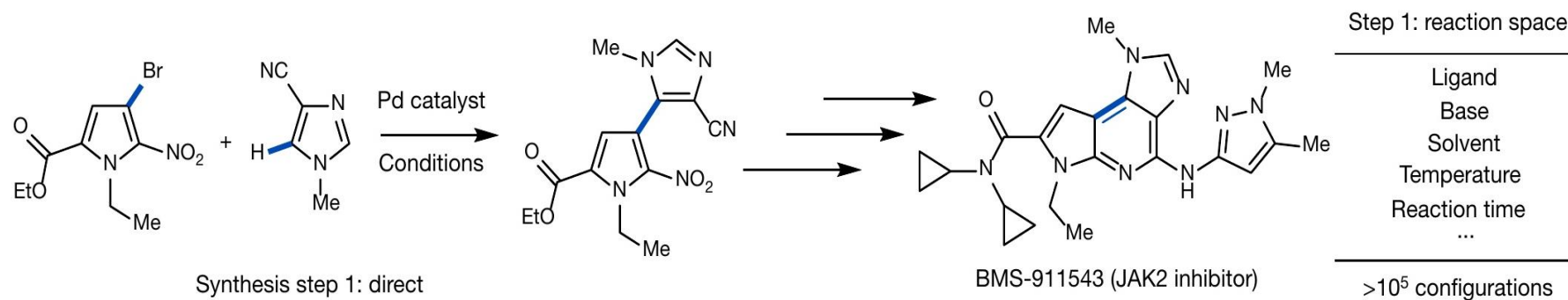
# Outline

- 1 · Optimizing when data acquisition is expensive**
- 2 · The Bayesian optimization loop
- 3 · Gaussian processes
- 4 · Categorical variables and noise
- 5 · Practical Scenarios: Multi-objective, batch

# Reaction Optimization

## Maximize the yield of a Suzuki coupling

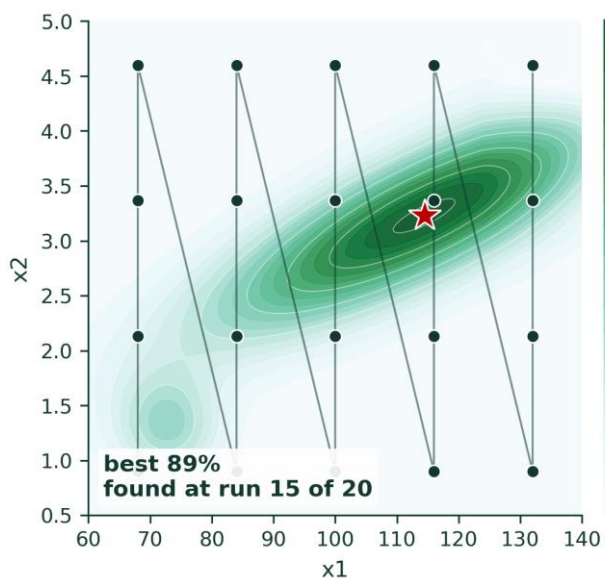
temperature · residence time · catalyst loading · ligand · base · solvent



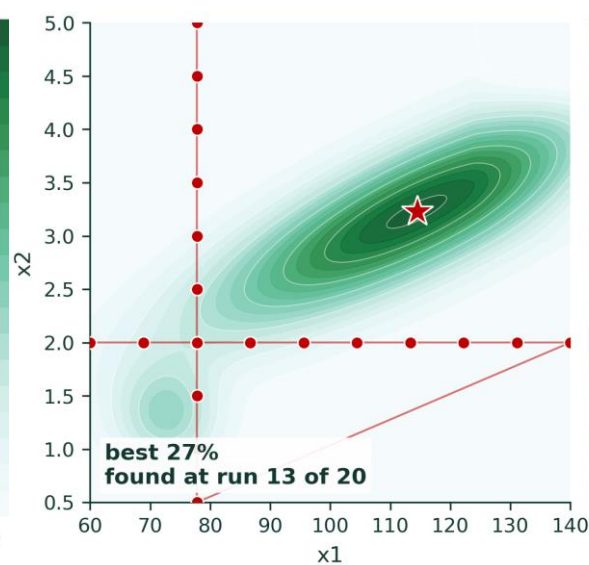
Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Martinez Alvarado, J. I.; Janey, J. M.; Adams, R. P.; Doyle, A. G. *Nature* **2021**, 590 (7844), 89–96.

# Baseline Approaches

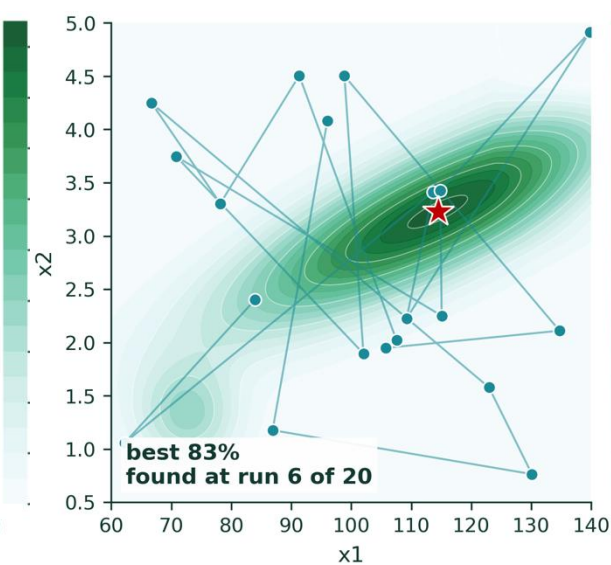
## Grid Search



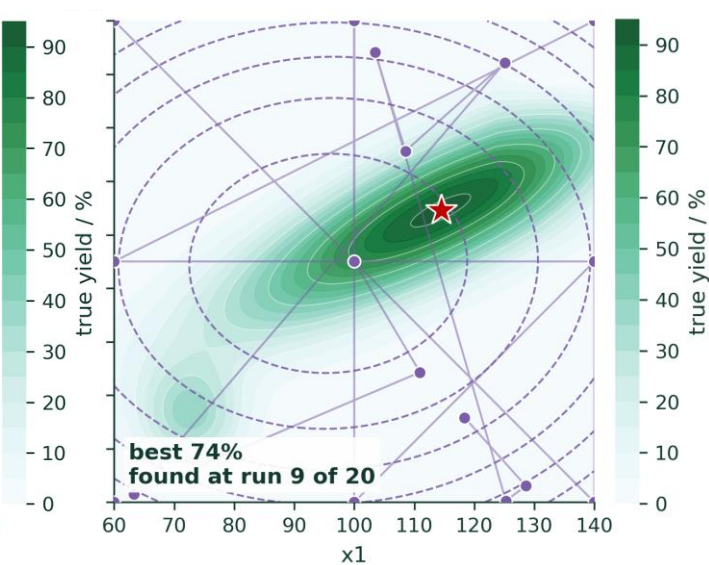
## OVAT



## Random



## DOE



Objective:  $\operatorname{argmax} f(x)$

$$x^* = \operatorname{argmax} f(x)$$

**Black-box**

**Non-convex, interacting**

**Mixed-variable**

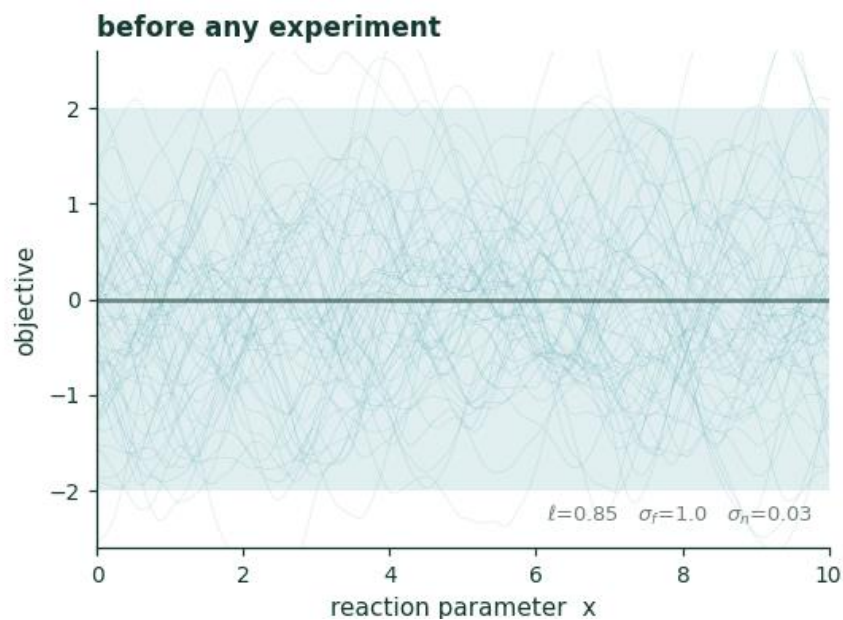
**Noisy**

**Expensive**

# Outline

- 1 · Optimizing when data acquisition is expensive
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# Bayesian Optimization



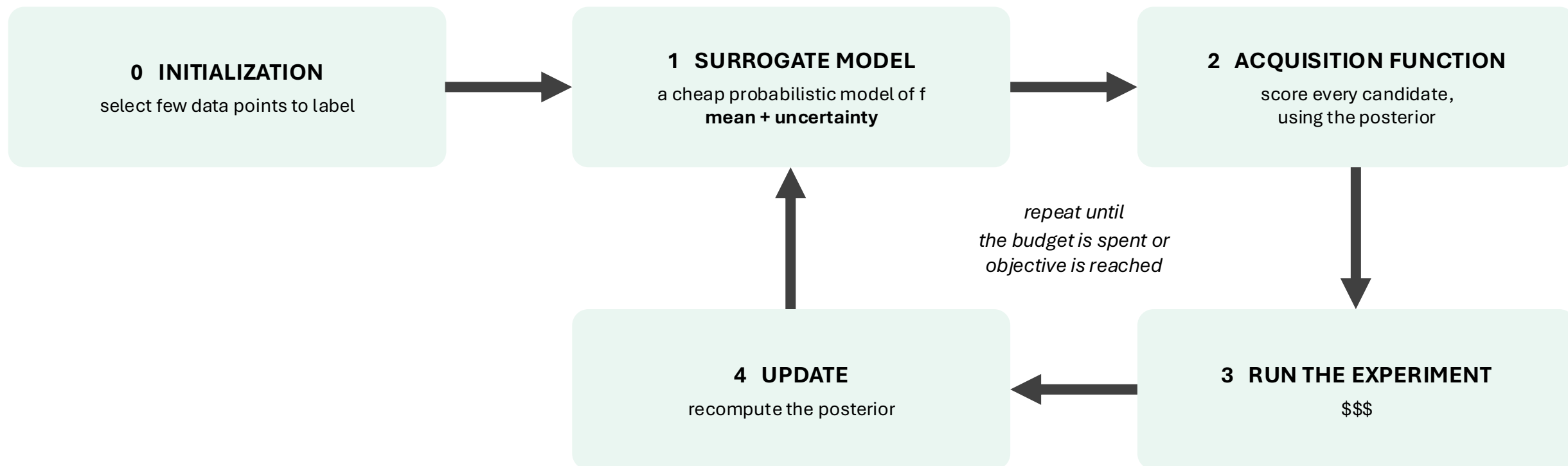
## Definition:

A sequential model-based strategy for optimization of black-box objective functions whose evaluations are costly.

The method constructs a probabilistic model of the unknown function and uses the resulting predictive distribution to choose the next evaluation point. This choice is made by optimizing a sampling criterion, also called an acquisition function.

[https://en.wikipedia.org/wiki/Bayesian\\_optimization](https://en.wikipedia.org/wiki/Bayesian_optimization)

# Bayesian Optimization Loop





# Bayes' law, stated over functions

$$p(f | D) \propto p(D | f) \cdot p(f)$$

## **PRIOR** $p(f)$

What functions are plausible before any data.

Not “yield is about 60%” but “yield varies smoothly with T, over tens of degrees”.

## **LIKELIHOOD** $p(D|f)$

How strongly the measurements constrain f.

Contains the experimental noise model.

Noiseless likelihood  $\Rightarrow$  the surrogate is forced through every point, including your bad ones.

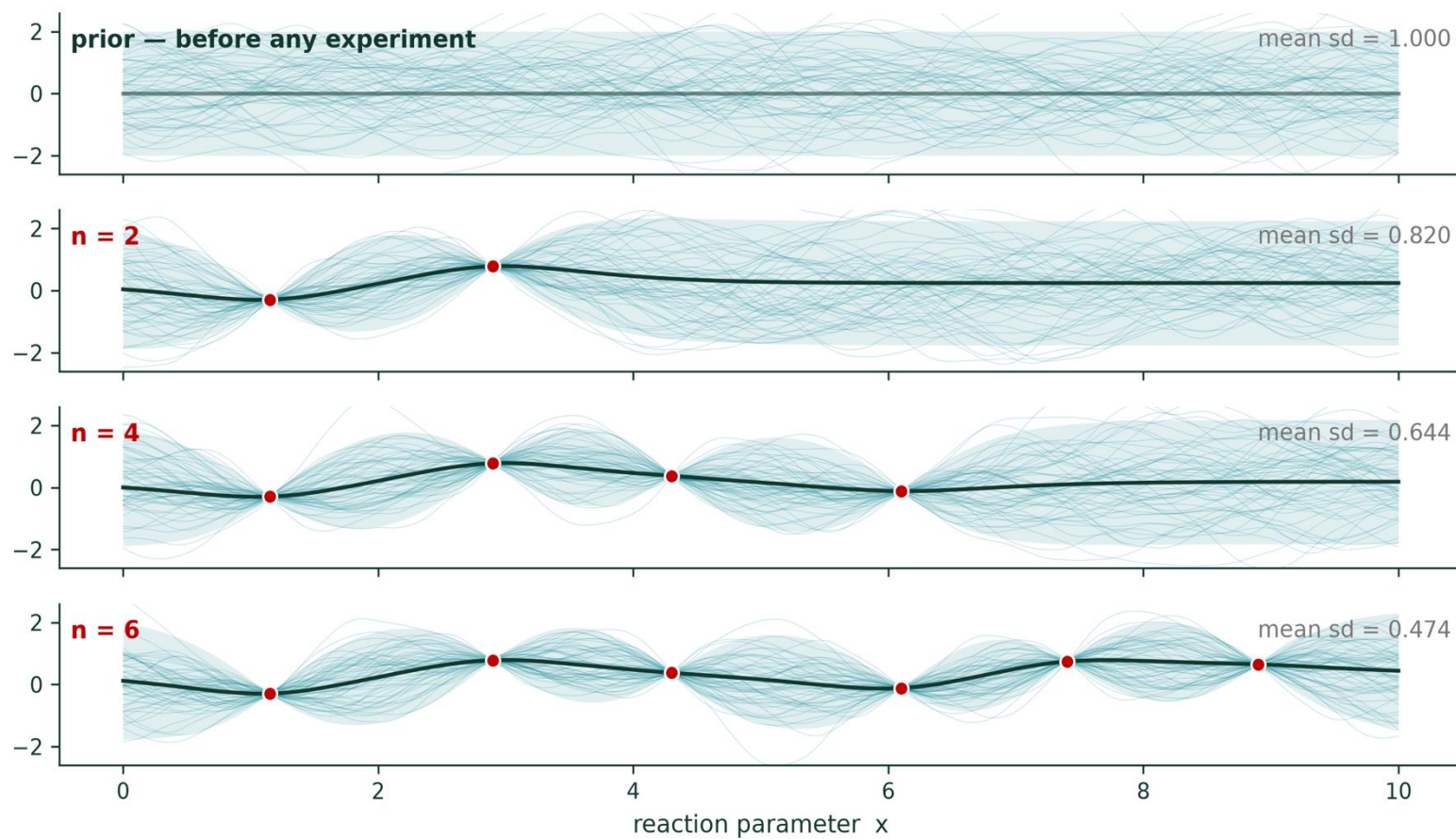
## **POSTERIOR** $p(f|D)$

The updated belief.

Narrow where you have measured.

Wide where you have not.

# The posterior is not a fit it is what survives



$$p(f)$$

Data  
acquisition

$$p(f | D)$$

# Decision Making

$$p(f(x) | D) = N(\mu(x), \sigma^2(x))$$

**Candidate A**

predicted 62%

$\sigma = 1\%$

**Candidate B**

predicted 55%

$\sigma = 15\%$

Which do you run next?

**Expected Mean Value + Model Uncertainty are needed for decision making**

## Closed Form

$$\mu(x) = k_x^T K^{-1} y \quad \sigma^2(x) = k_{xx} - k_x^T K^{-1} k_x$$

*with  $K = K(X, X) + \sigma_n^2 I$*

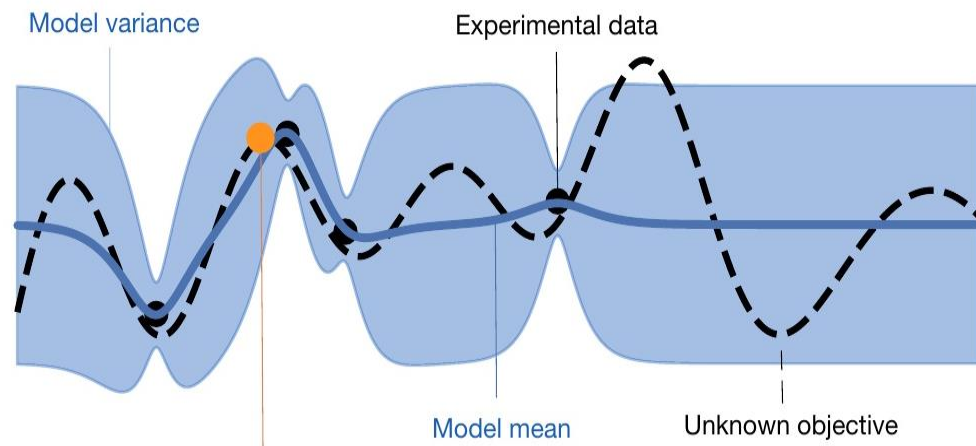
**$\sigma^2$  does not contain  $y$ .**

Where you sampled sets the uncertainty — not what you measured.

If you only consider uncertainty, you can plan a campaign before running it. This is the seed of tomorrow's Active Learning lecture.

**True only at fixed hyperparameters ( $\theta$ ). Once  $\ell$ ,  $\sigma_f$ ,  $\sigma_n$  are fitted to the data,  $y$  re-enters through  $\theta$ .**

## Reminder on the Objective

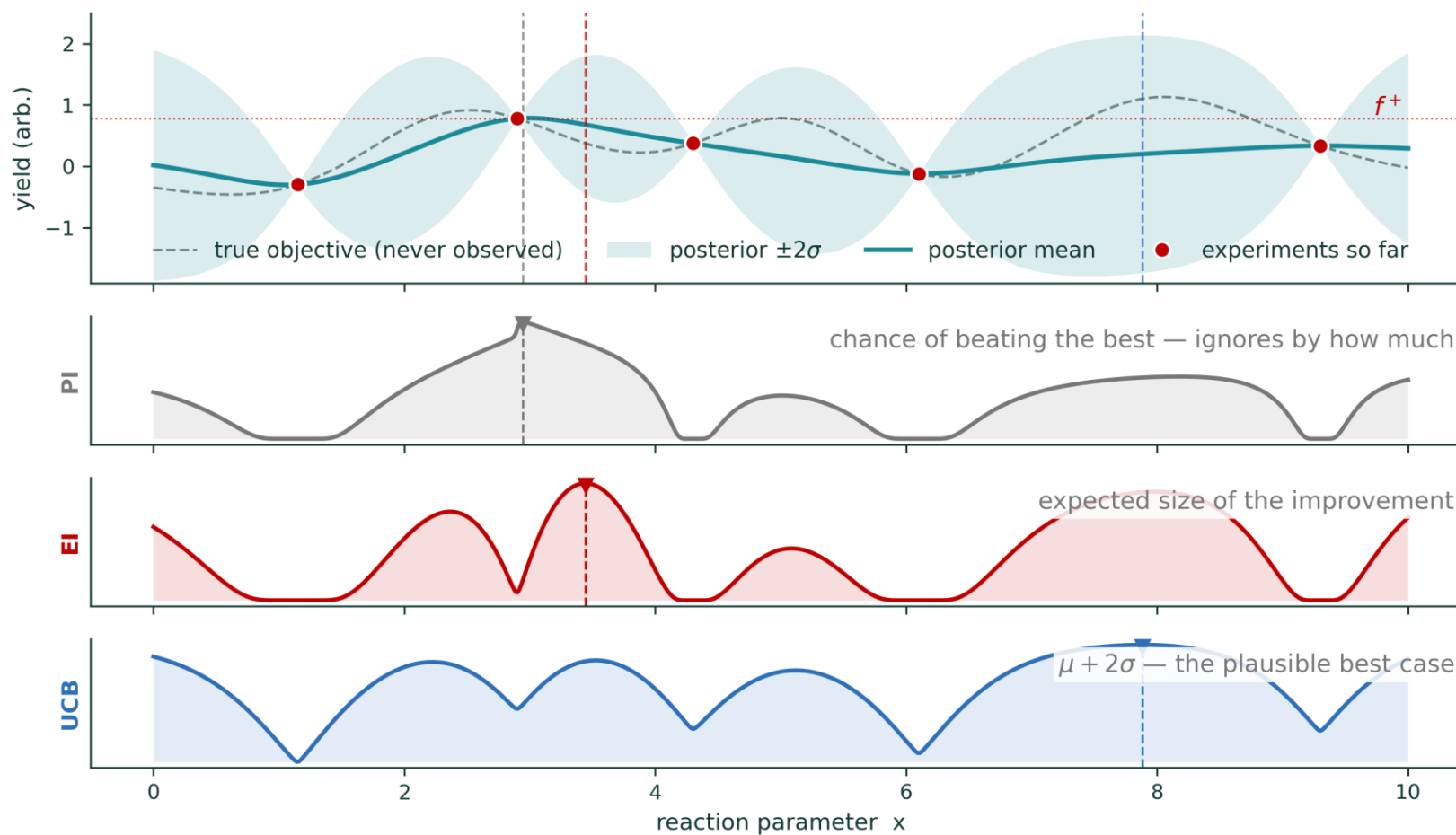


$$x^* = \operatorname{argmax} f(x)$$

We cannot optimize  $f$ , too expensive.  
**Instead optimize a score for how useful an experiment would be**

# Decision Making/Acquisition Function using the Posterior

PI: Probability of Improvement  
EI: Expected Improvement  
UCB: Upper Confidence Bond



# Expected Improvement (EI)

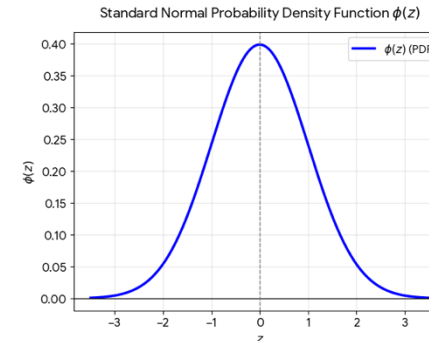
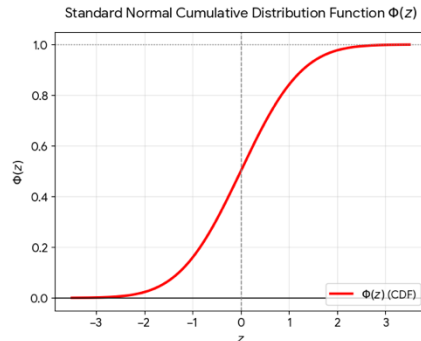
$$\alpha\_EI(x) = E[\max(0, f(x) - f^*)] = (\mu - f^*)\Phi(z) + \sigma\phi(z)$$
$$z = (\mu - f^*) / \sigma$$

$$(\mu - f^*) \Phi(z)$$

large where the mean is high  $\rightarrow$  EXPLOITATION

$$\sigma \phi(z)$$

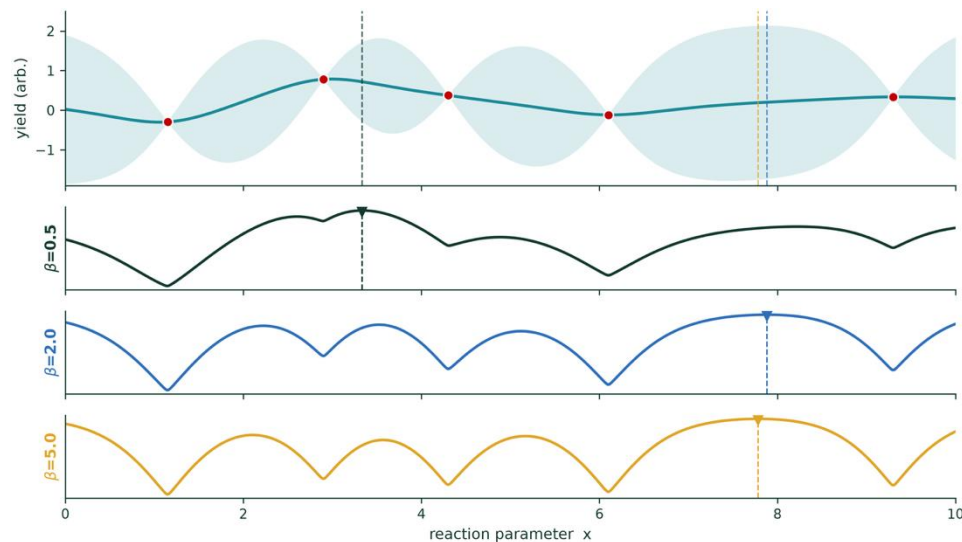
grows with  $\sigma \rightarrow$  EXPLORATION



Note:  $\Phi(z)$  alone is Probability of Improvement (PI). The EGO paper comes out of the design-of-computer-experiments tradition — the same lineage as DOE.  
Jones, D. R.; Schonlau, M.; Welch, W. J. *J. Global Optim.* **1998**, 13 (4), 455–492.

# Upper Confidence Bound (UCB)

$$\alpha_{UCB}(x) = \mu(x) + \beta \cdot \sigma(x)$$



$\beta$  small  $\rightarrow$  greedy

$\beta$  large  $\rightarrow$  explorer

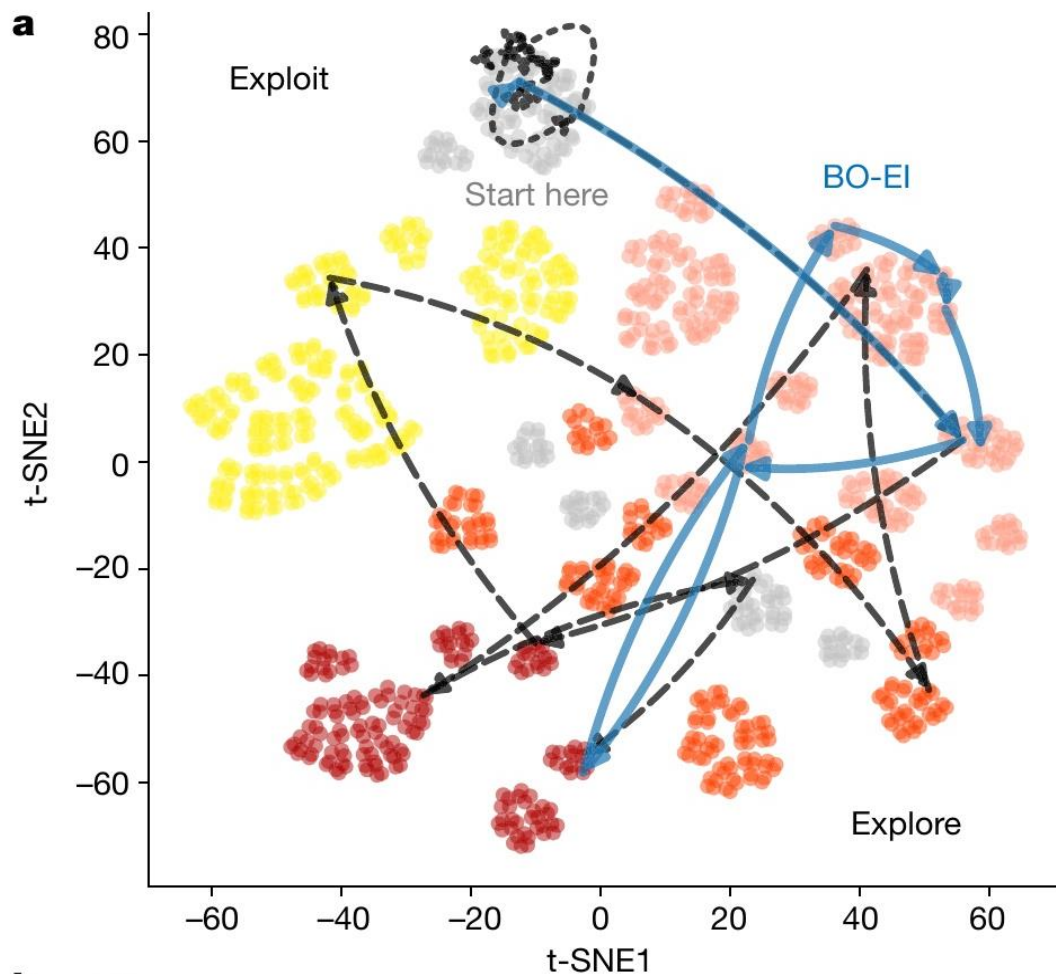
$\beta = 2$  is the usual default

fig\_06\_ucb\_beta\_sweep.ipynb

Srinivas, N.; Krause, A.; Kakade, S. M.; Seeger, M. W. *IEEE Trans. Inf. Theory* **2012**, 58 (5), 3250–3265.



# Exploration – Exploitation Trade-off



→ Pure EXPLOIT stays in the cluster it started in.

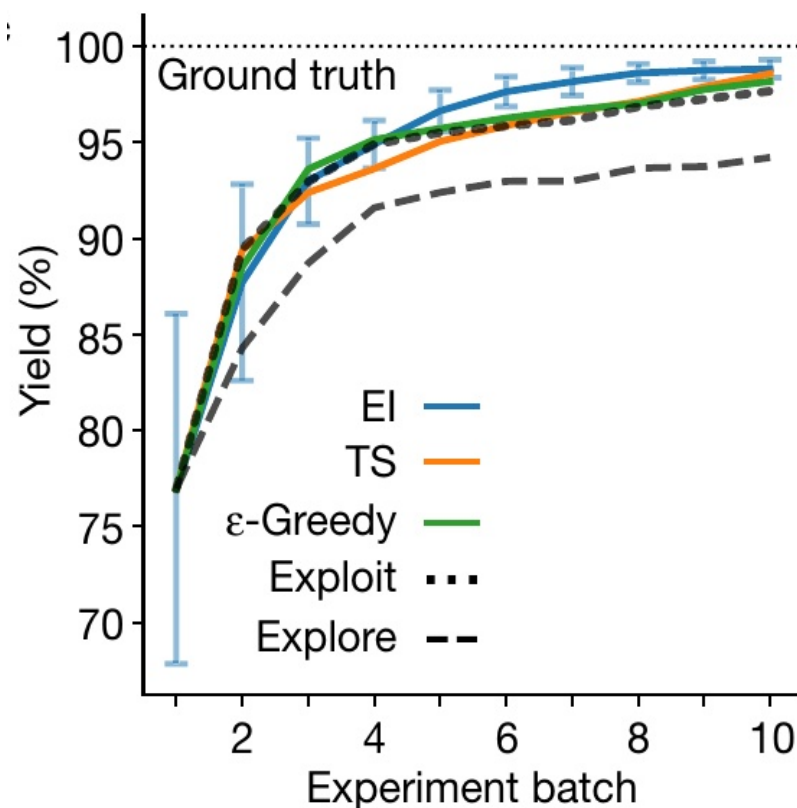
→ Pure EXPLORE tours the whole space

→ **BO with EI visits three of the five clusters, then commits.**

Fig. 3a.

Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Martinez Alvarado, J. I.; Janey, J. M.; Adams, R. P.; Doyle, A. G. *Nature* **2021**, 590 (7844), 89–96.

# The Acquisition Function Matters Less Than You Think



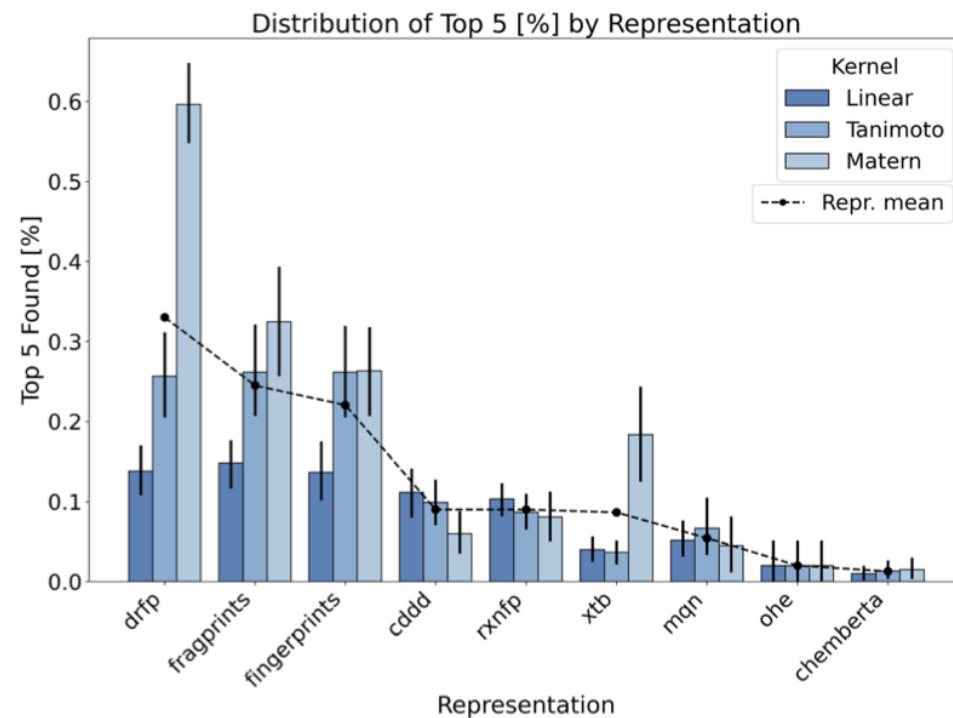
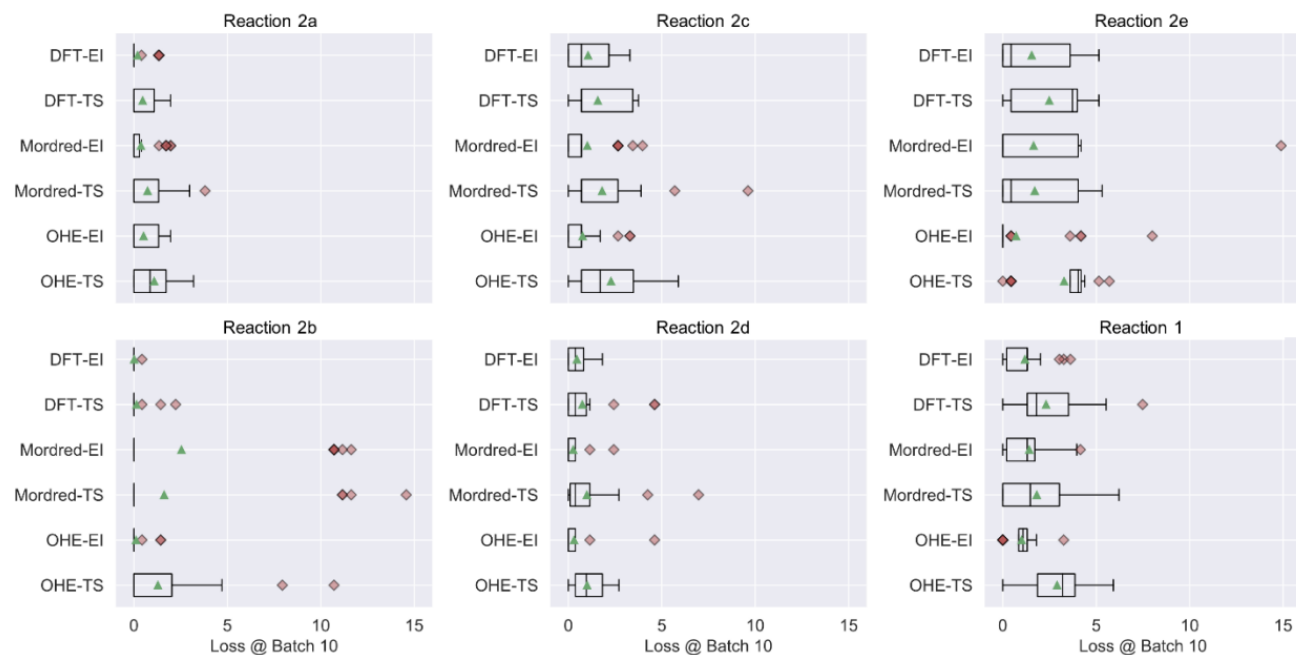
**EI, Thompson sampling and  $\epsilon$ -greedy all converge together.**

Pure exploit and pure explore are the only ones that lose.

Fig. 3c,f — average of 30 runs, error bars  $0.5\sigma$ .

Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Martinez Alvarado, J. I.; Janey, J. M.; Adams, R. P.; Doyle, A. G. *Nature* **2021**, 590 (7844), 89–96.

# Role of the Representation and Kernel

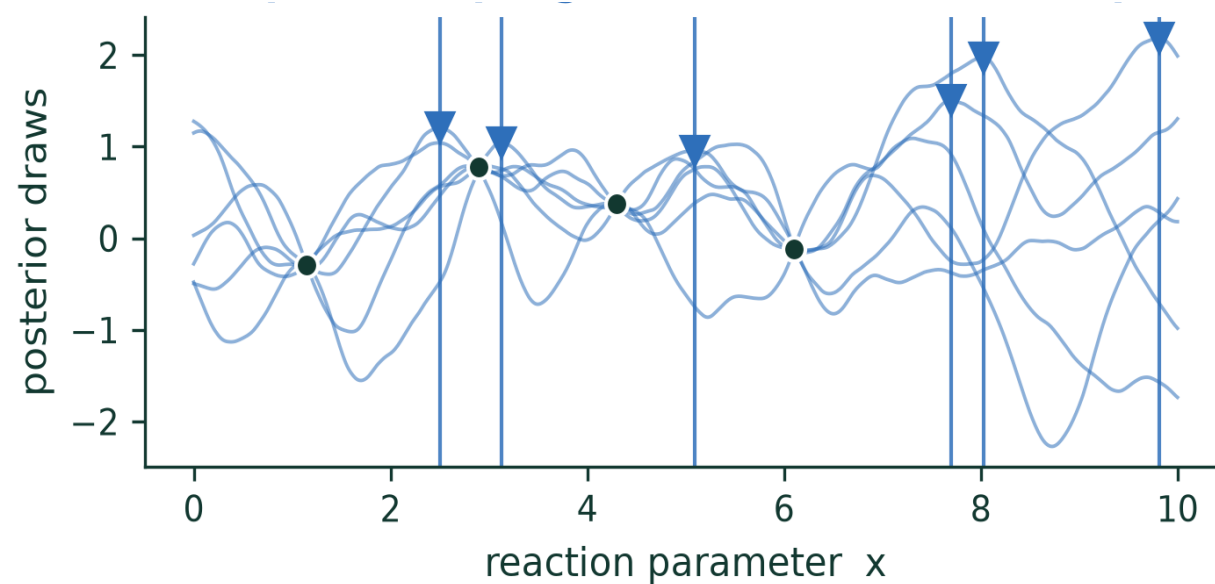


Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Martinez Alvarado, J. I.; Janey, J. M.; Adams, R. P.; Doyle, A. G. *Nature* **2021**, 590 (7844), 89–96.

Ranković, B.; Griffiths, R.-R.; Moss, H. B.; Schwaller, P. *Digital Discovery* **2024**, 3 (4), 654–666.

# Thompson Sampling

Draw one function from the posterior. Select  $x$  that maximizes that draw as a next experiment



# Plan

- 1 · Optimizing when data acquisition is expensive
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- 3 · Gaussian processes**
- 4 · Categorical variables and noise
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# Kernel

**If two conditions are similar, how similar should their outcomes be?**

RBF

$$k(x, x') = \sigma_f^2 \exp(-\|x - x'\|^2 / 2\ell^2)$$

MATÉRN 5/2

*same idea, rougher assumption about  $f$*

RBF implies  $f$  is infinitely differentiable — a strong claim about a surface with phase boundaries and catalyst death. Matérn 5/2 assumes only twice-differentiable.

# Hyperparameters

RBF kernel

$$k(x, x') = \sigma_f^2 \exp(-\|x - x'\|^2 / 2\ell^2)$$

## $\ell$ length-scale

“How far must I move in temperature before the yield changes?”

$\ell = 10\text{ }^\circ\text{C}$  is a claim you can sanity-check.

Too small  $\rightarrow$  reverts to the mean between points; BO degenerates to random search.

Too large  $\rightarrow$  oversmooths, misses a narrow optimum.

## $\sigma_f^2$ signal variance

“How much does yield vary across the whole space?”

Sets the vertical scale of the prior.

Too small  $\rightarrow$  over-confident, under-explores.

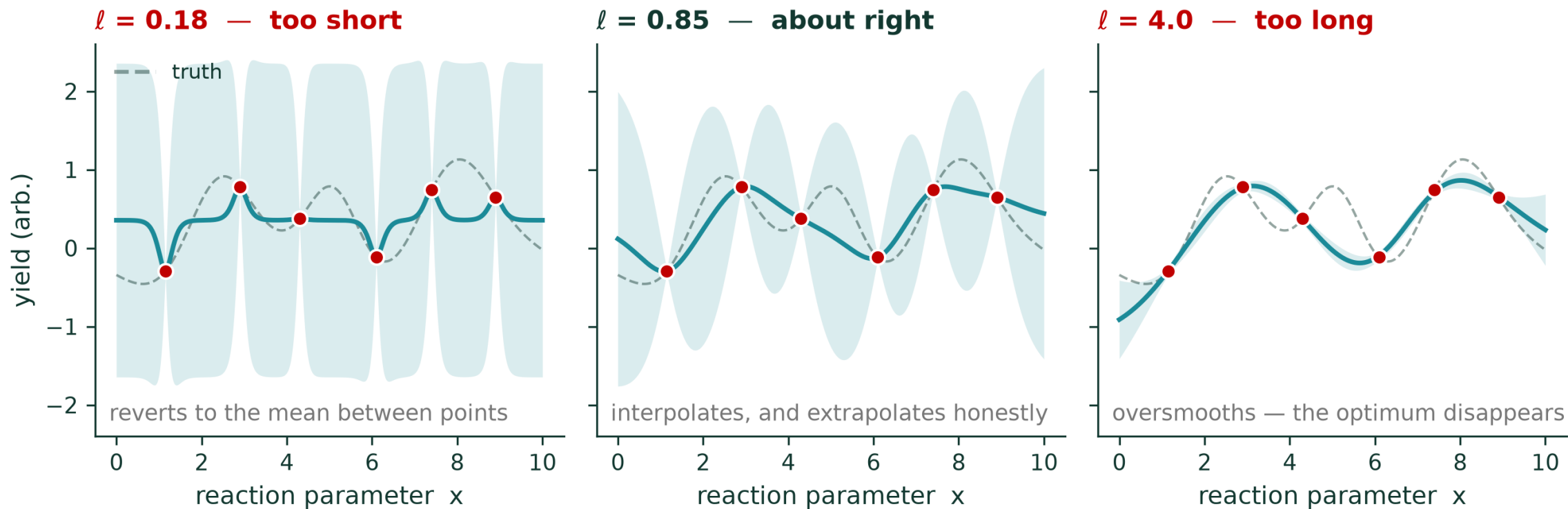
## $\sigma_n^2$ noise

Your experimental error. The replicate error of your assay.

Set to  $\sim 0$  and the GP interpolates every point exactly — including one bad measurement — and BO chases a lucky outlier for the rest of the campaign.

The single most common practical error in chemistry BO.

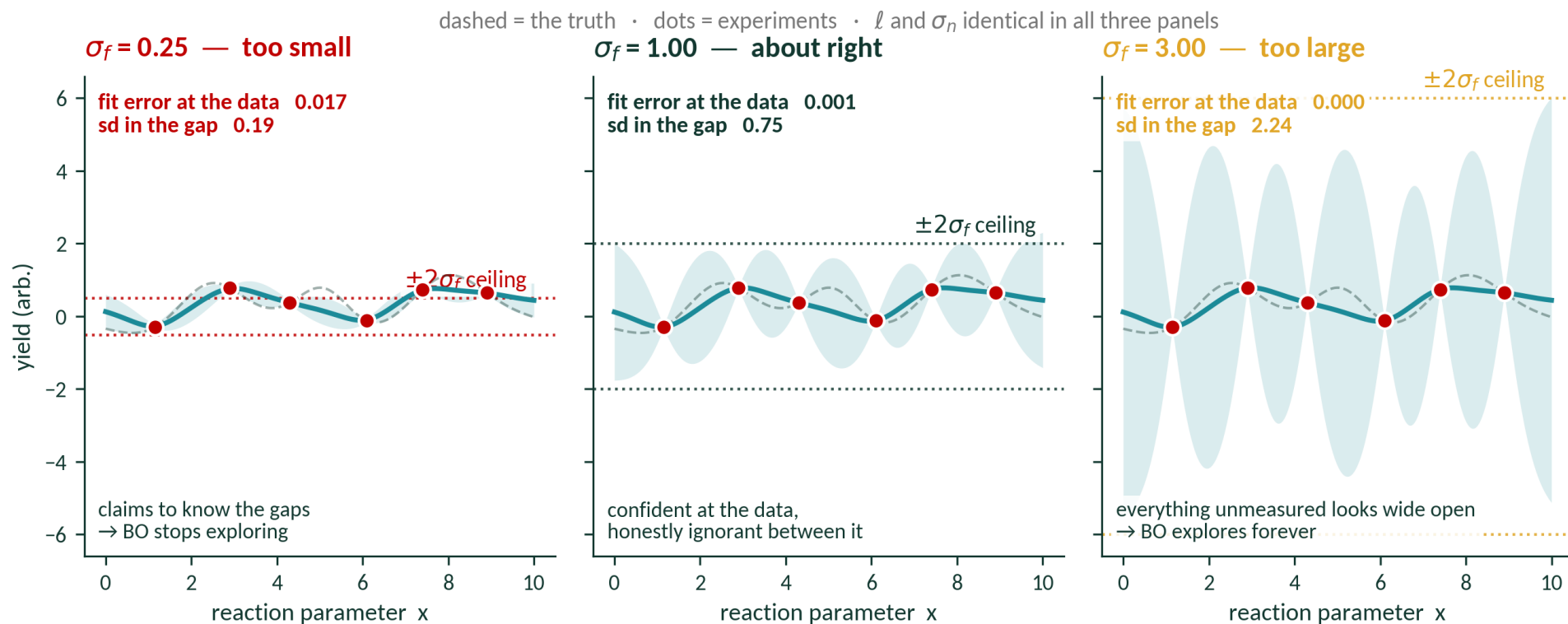
# Length Scale



Too short and the posterior snaps back to the mean between points — BO degenerates towards random search. Too long and the optimum is smoothed away.

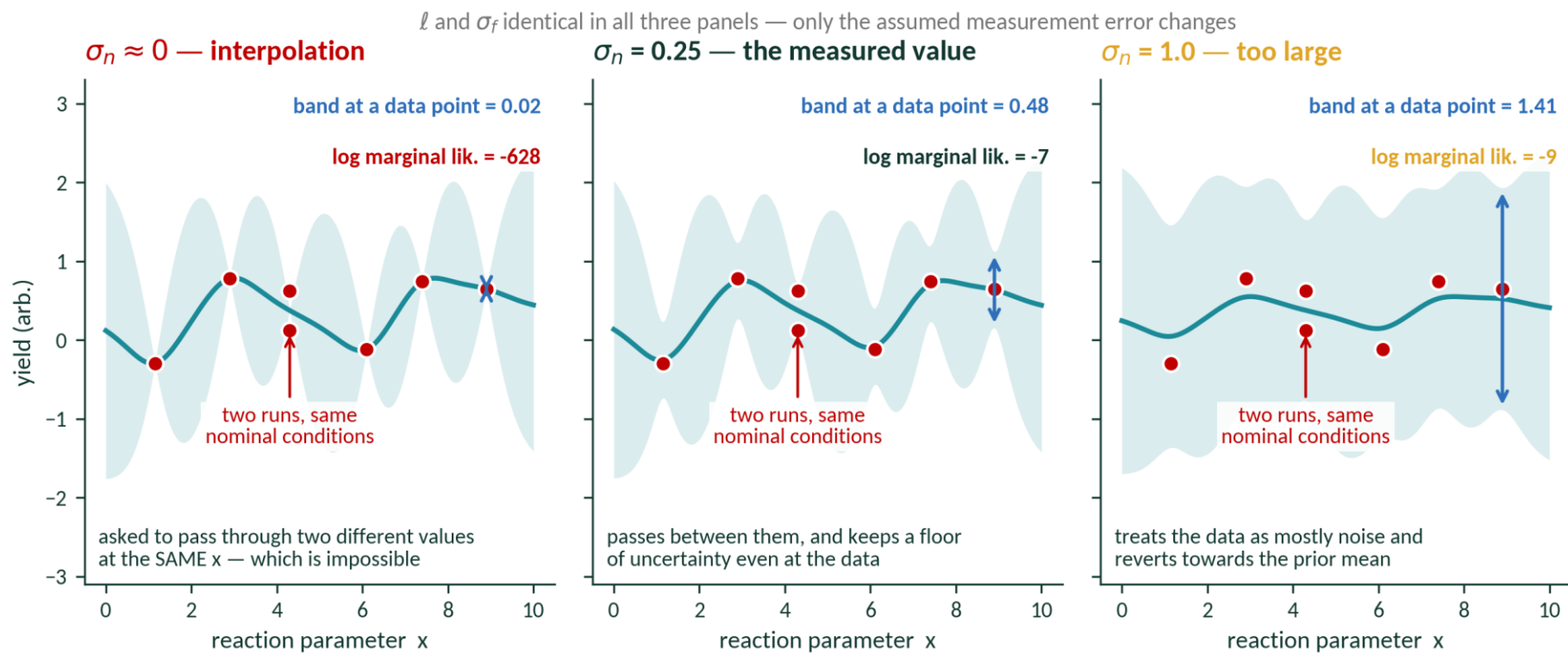


# $\sigma_f$ — Signal Variance



All three fit the data equally well.  $\sigma_f$  changes nothing at the observations — it sets the CEILING of the band in the gaps, and therefore how much BO wants to explore.

# $\sigma_n$ — Data Noise



fig\_19\_sigma\_n.ipynb ·  $\ell$  and  $\sigma_f$  identical; one pair of runs is a replicate

# Hyperparameter Fitting

$$\log p(y | X, \theta) = -\frac{1}{2} y^T K^{-1} y - \frac{1}{2} \log |K| - (n/2) \log 2\pi$$

$$-\frac{1}{2} y^T K^{-1} y$$

data fit — rewards explaining the observations

$$-\frac{1}{2} \log |K|$$

complexity penalty — punishes over-flexible models

# Plan

- 1 · Optimizing when data acquisition is expensive
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- 5 · Practical Scenarios: Multi-objective, batch

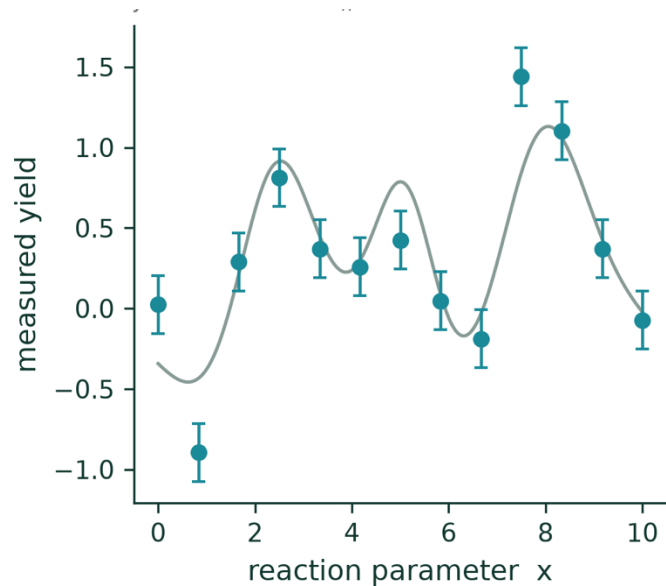
# Three different things get called “noise”

## HOMOSCEDASTIC

constant  $\sigma_n$  across the space

assay error roughly the same everywhere

→ *standard GP* — just don't set  $\sigma_n \approx 0$

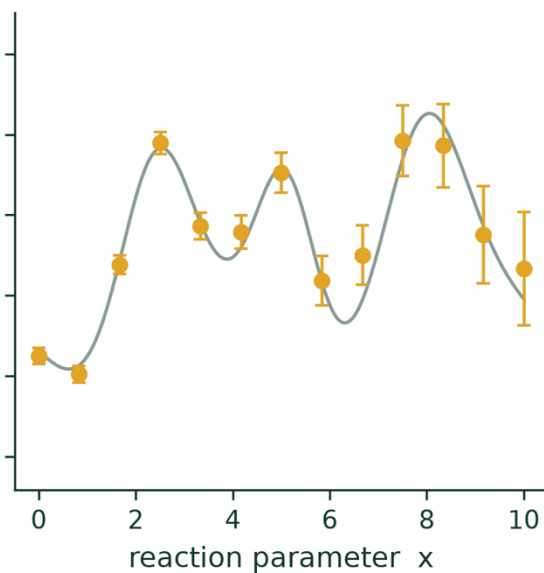


## HETEROSCEDASTIC

$\sigma_n$  varies with x

variance blows up at high T where the catalyst dies; low-conversion regions are proportionally noisier

→ *heteroscedastic GP*

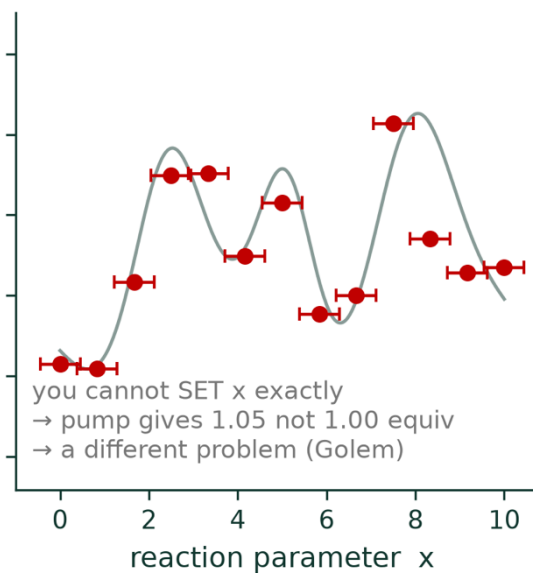


## INPUT NOISE

you cannot SET x exactly

the pump delivers 1.05 not 1.00 equiv; the reactor overshoots by 3 °C

→ *Golem* — a different problem entirely



## Why noise biases BO, rather than just slowing it down

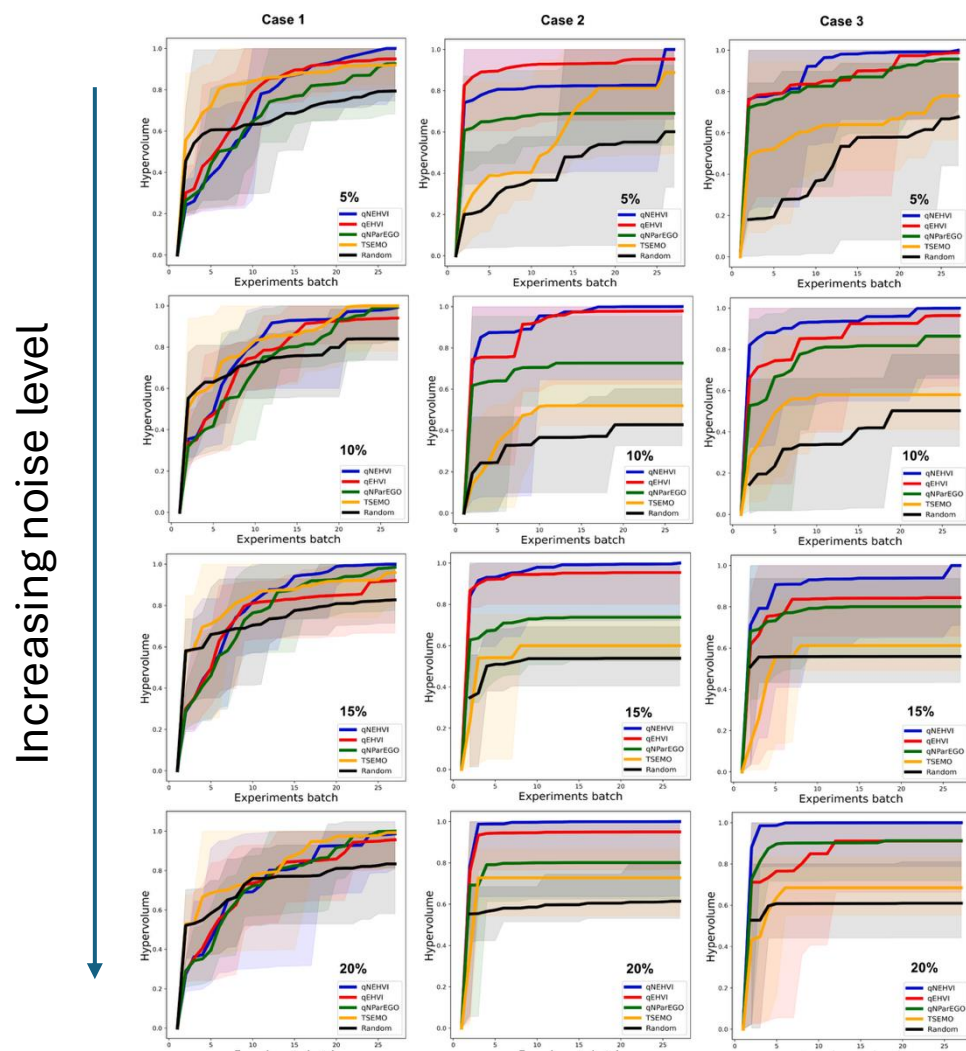
El is defined against  $f^+$ , the best value observed so far — and treats it as truth.

Under noise,  $f^+$  is probably a lucky measurement.

If El anchors to a number that is too high, it under explores, and spends the budget chasing an outlier.

**Standard El trusts your best data point. Noisy El does not.**

# Published: which acquisition function survives the noise



## qNEHVI — noisy expected hypervolume improvement

Integrates over the posterior at every point already run, so the current Pareto front is treated as uncertain rather than as fact. Top of all twelve panels.

## qEHVI — expected hypervolume improvement

Same hypervolume reward, but conditions on the observed front as if it were exact. Level with qNEHVI at 5 % noise; falls behind as noise grows.

## qNParEGO — random-weight scalarisation

Draws a fresh weight vector per batch point, collapses the objectives with an augmented Tchebycheff function, then runs noisy EI. Cheap.

## TSEMO — Thompson sampling, multi-objective

NSGA-II over random Fourier feature draws of the GPs. Strong at 5 %, degrades most once noise reaches 10 %.

## Random

The control. Note how close it gets in Case 1 — that panel is a wash.

# Your real variables are not numbers

What is the distance between  $\text{PPh}_3$  and XPhos? Between DMF and toluene?

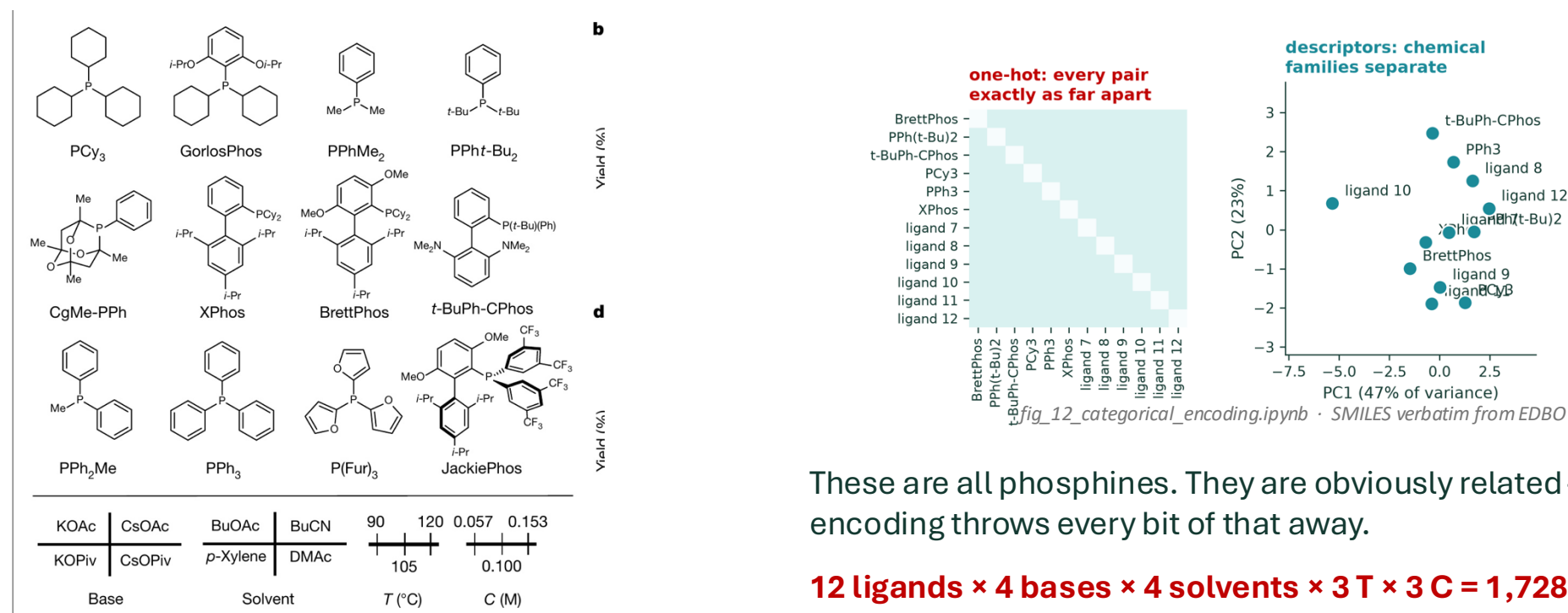
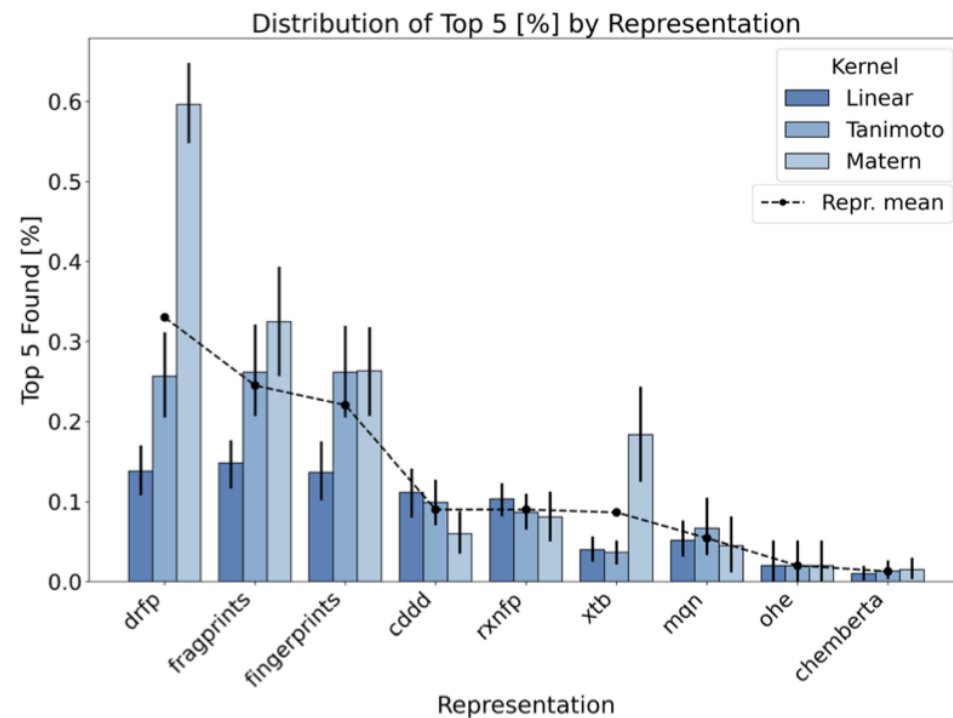
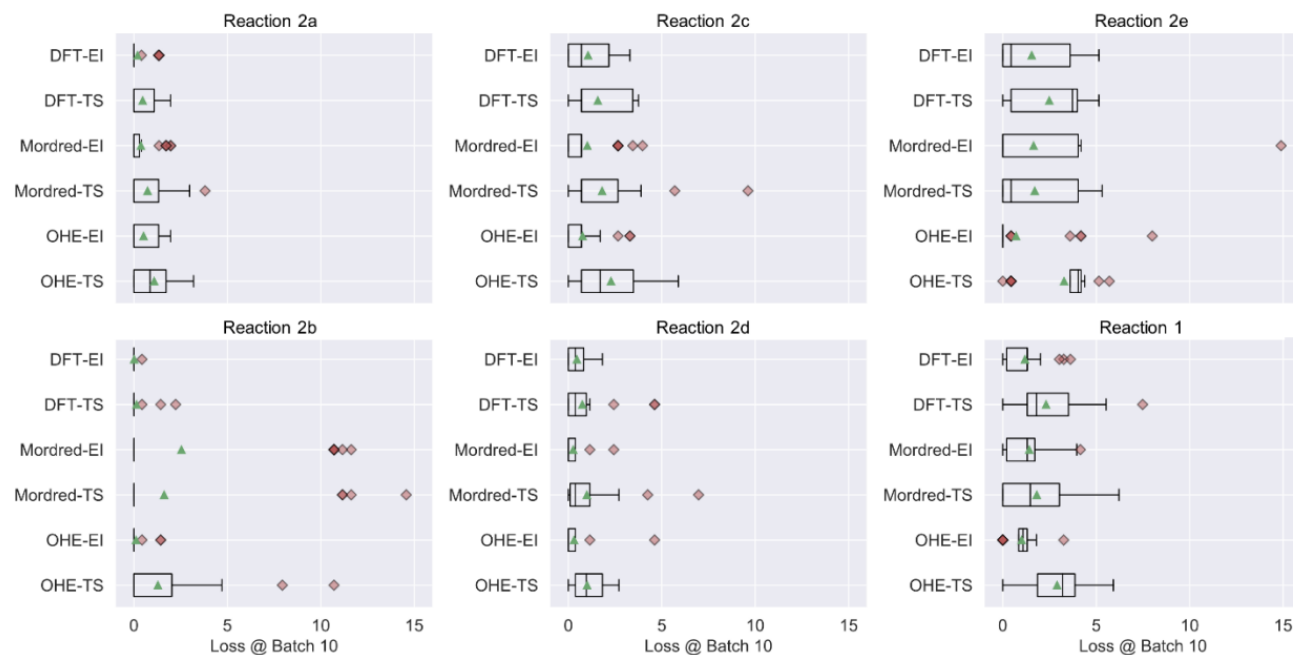


Fig. 4a.

Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Martinez Alvarado, J. I.; Janey, J. M.; Adams, R. P.; Doyle, A. G. *Nature* **2021**, 590 (7844), 89–96.



# Role of the Representation and Kernel



Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Martinez Alvarado, J. I.; Janey, J. M.; Adams, R. P.; Doyle, A. G. *Nature* **2021**, 590 (7844), 89–96.

Ranković, B.; Griffiths, R.-R.; Moss, H. B.; Schwaller, P. *Digital Discovery* **2024**, 3 (4), 654–666.

# DOE vs BO | When to choose which ?

|                                                                                   | DOE (factorial + RSM)                        | Bayesian optimization                              |
|-----------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|
| Design                                                                            | fixed, chosen up front, all at once          | sequential, adapts after every result              |
| Model                                                                             | polynomial of fixed form                     | non-parametric; flexibility grows with data        |
| Uncertainty                                                                       | CI's on coefficients                         | full posterior at every point                      |
| Primary goal                                                                      | UNDERSTAND — main effects, interactions      | FIND THE ARGMAX                                    |
| Categorical                                                                       | levels; combinatorially expensive            | descriptors let information transfer               |
| Best when                                                                         | few factors, cheapish runs, you want a model | expensive runs, nonlinear, you want the best point |
| Complementary: screening DOE to drop irrelevant factors, then BO on what is left. |                                              |                                                    |

# Plan

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# Multi-Objective

## Process

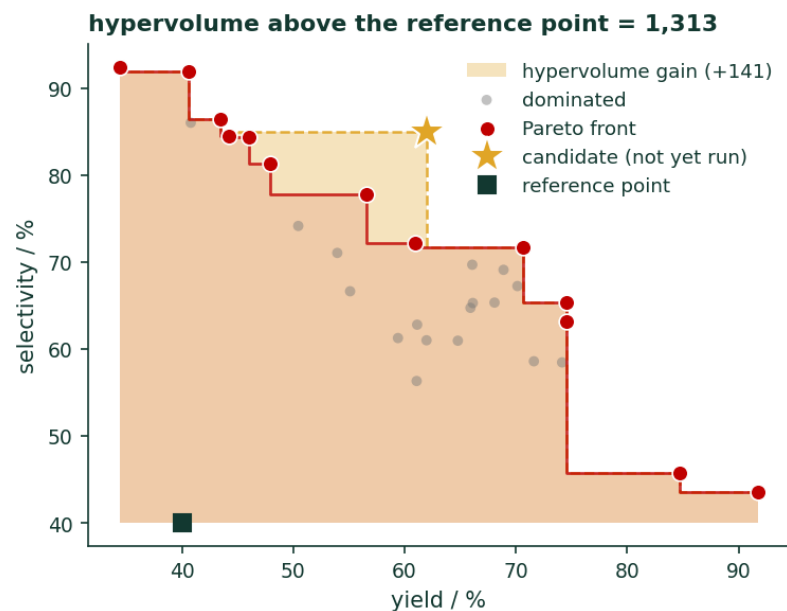
yield AND selectivity AND cost AND catalyst loading

## Medicinal

potency AND solubility AND metabolic stability AND low hERG

## Materials

conductivity AND stability AND processability

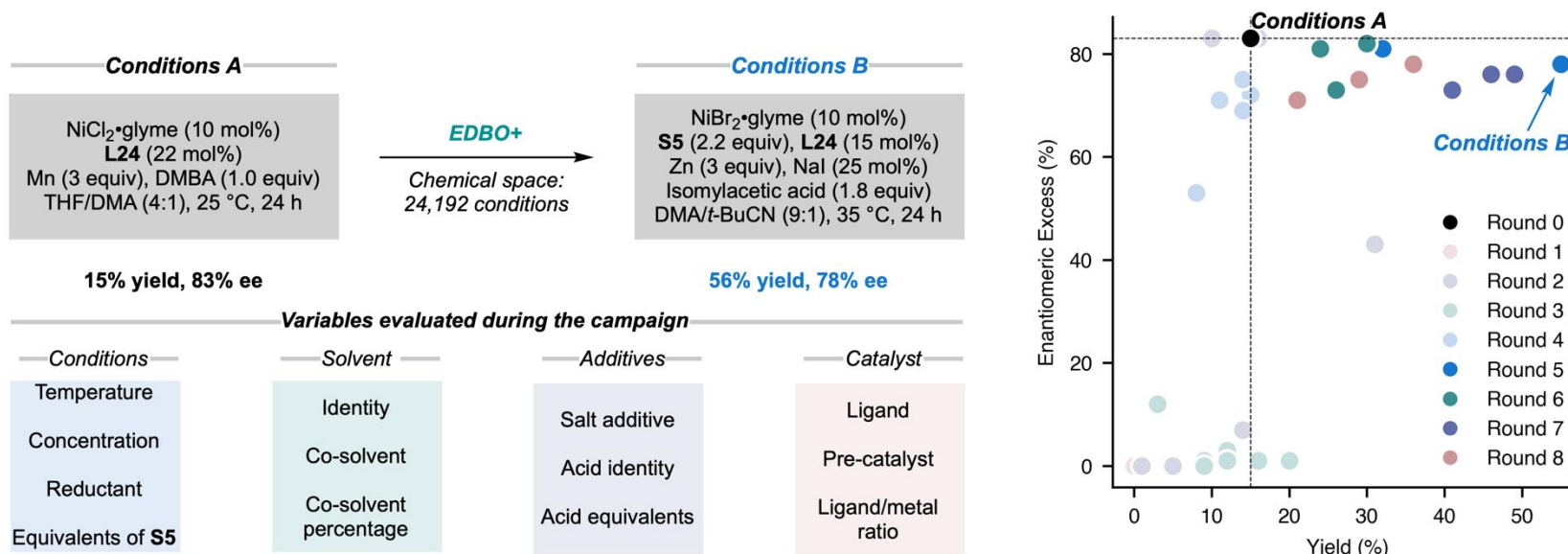


**EHVI : expected hypervolume improvement.**

**“How much do I expect this experiment to grow the frontier?”**

fig\_16\_pareto\_front.ipynb

# Practical take away: Tells you when you can/should stop



## Condition space update after optimization plateau

### Rounds 1 – 4:

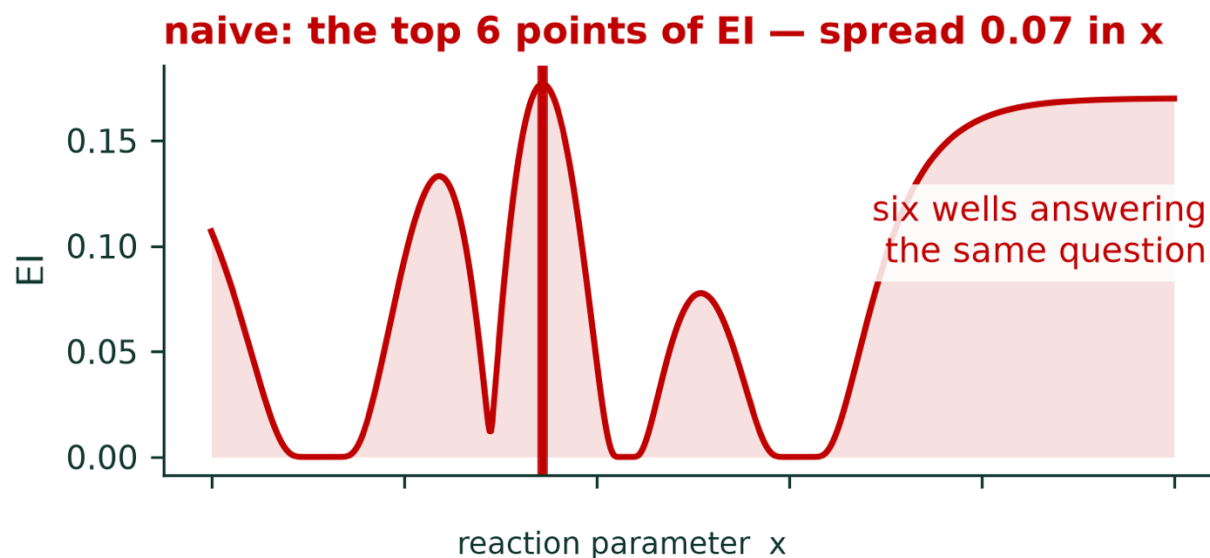
- Solvent
- Additive
- Reductant
- Ligand
- Precatalyst
- L/M ratio
- Concentration

### Rounds 5 – 8:

- Co-solvent
- Co-solvent %
- Concentration
- Acid identity
- Acid equivalent
- **S5** equivalents

# Batching

Take the top  $q$  points of  $\alpha(x)$  and they cluster — eight wells answering the same question.



## q-EI

joint expected improvement of a SET of  $q$  points. Principled, pricier, standard in BoTorch.

## Thompson sampling

draw  $q$  posterior functions, maximise each. Diversity is automatic — you planted this earlier.

## Greedy + fake data

constant liar inserts a FIXED constant; Kriging believer inserts the MODEL'S prediction. Not the same thing.

## Local penalization

no fabricated data at all — multiply  $\alpha$  by a penalty around each point already chosen.

*With parallelism you stop minimizing the NUMBER of experiments and start minimizing wall-clock time or cost — and those give different designs.*

# q-El, formally

A batch is one decision about q points

$$\alpha_{qEI}(X) = E[ \max_i \max(0, f(x_i) - f^*) ] \quad X = \{x_1, \dots, x_q\}$$

## Why not just take the top q of EI?

Sum of individual EIs is maximized by q copies of the same point. The inner max pays the batch ONCE however many of its members improve, so redundancy earns nothing and diversity is automatic.

## What it costs

You now optimize jointly over q·d dimensions, and each evaluation is an N-sample average. This is why sequential-greedy approximations exist — and why BoTorch ships both.

## Under noise: q-NEI

$f^*$  is replaced by the max of posterior samples at the points already run. Same fix as slide 30, now inside a batch. qNEI → qNEHVI is the multi-objective version on slide 32.

Ginsbourger, D.; Le Riche, R.; Carraro, L. In *Computational Intelligence in Expensive Optimization Problems*; Tenne, Y., Goh, C.-K., Eds.; Springer: Berlin, 2010; pp 131–162.

Wang, J.; Clark, S. C.; Liu, E.; Frazier, P. I. *Oper. Res.* **2020**, 68 (6), 1850–1865.

Balandat, M.; Karrer, B.; Jiang, D. R.; Daulton, S.; Letham, B.; Wilson, A. G.; Bakshy, E. In *Advances in Neural Information Processing Systems 33*; Curran Associates: Red Hook, NY, 2020; pp 21524–21538.

# You will hit at least two of these

## Too few seed points

BO with 3 initial points is not BO. ~5–10 per dimension, or a DOE screen.

## Bounds set wrong

The optimum is at 150 °C; you bounded at 120. It converges beautifully to the wrong answer.

## Noise ignored

$\sigma_n \approx 0 \rightarrow$  interpolates a lucky replicate  $\rightarrow$  15 experiments revisiting it.

## Dimension too high

Degrades toward random search without structure. Reduce the space first — that is what a screening DOE is for.

## Deceptive landscapes

Phase changes, precipitation, catalyst death. The smoothness prior is simply wrong across the discontinuity.

## Badly posed objective

Optimise yield alone  $\rightarrow$  99% yield at 200 equivalents and 72 hours. It optimises exactly what you asked.

## Robustness

The sharp optimum may not survive process variation. A lower, broader one is often the better manufacturing answer.

Robustness (Golem). Aldeghi, M.; Häse, F.; Hickman, R. J.; Tamblyn, I.; Aspuru-Guzik, A. *Chem. Sci.* **2021**, 12 (44), 14792–14807.

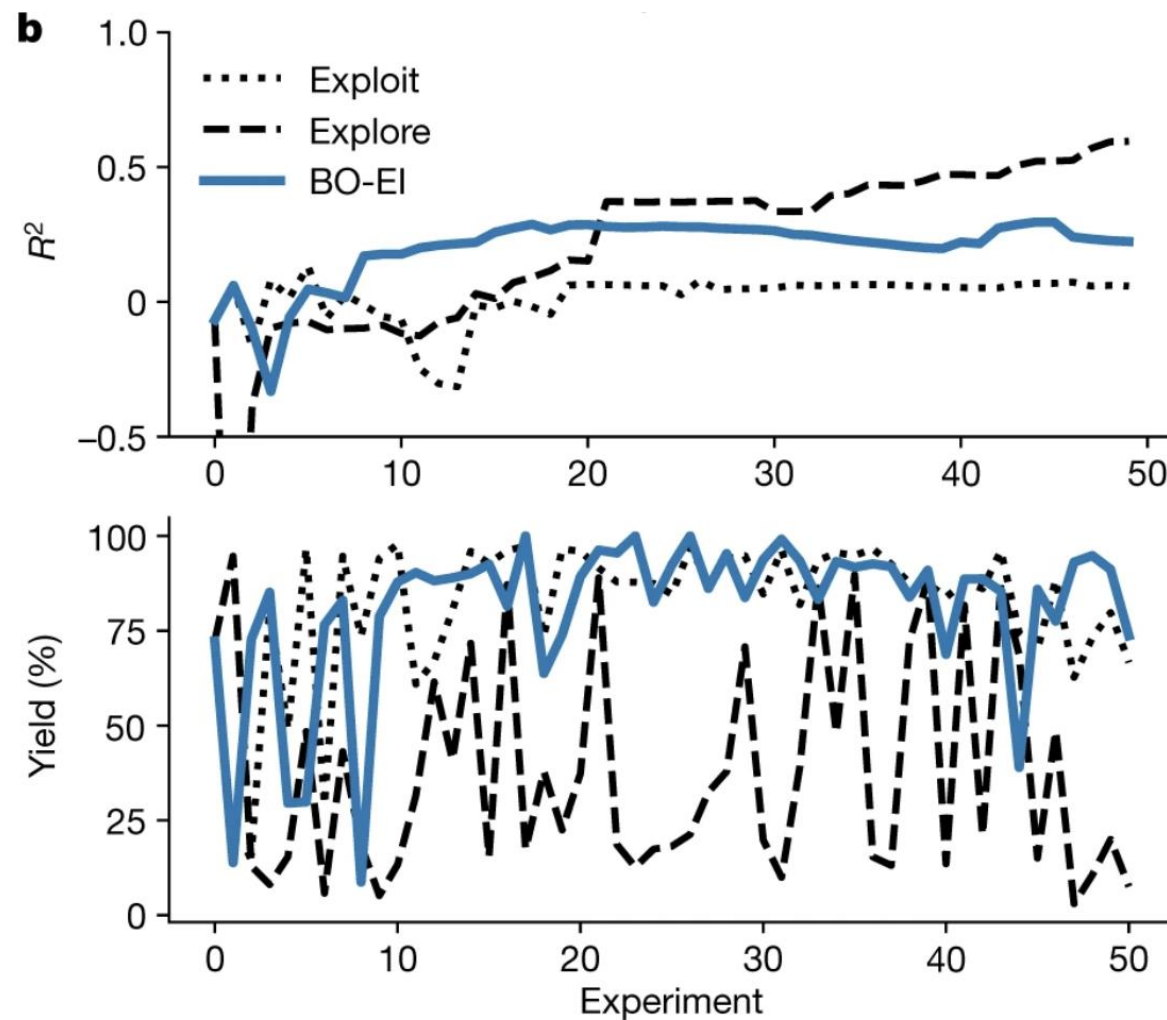


# The framework does not need a Gaussian process

| SURROGATE                       | UNCERTAINTY FROM             | GOOD WHEN                                      |
|---------------------------------|------------------------------|------------------------------------------------|
| <b>Gaussian process</b>         | closed-form posterior        | $n \lesssim 10^3$ , continuous, low-d          |
| <b>Random forest</b>            | variance across trees        | categorical / conditional spaces; used in SMAC |
| <b>NN ensembles, MC dropout</b> | disagreement between members | $n \gg 10^3$ , structured inputs — MolPAL      |

**Stay with a GP until experiment number exceeds a few thousand, or your inputs stop being vectors.**

## Where this goes next



Gutiérrez-Valencia, N. E.; Schleinitz, J.; Wild, T. H.; Williams, W. L.; Martin, D.; Doyle, A. G.; Reisman, S. E.; Sigman, M. S. *ACS Catal.* **2026**. <https://doi.org/10.1021/acscatal.6c02585>.

# References

## METHOD

Jones, D. R.; Schonlau, M.; Welch, W. J. *J. Global Optim.* **1998**, 13 (4), 455–492. (EGO / expected improvement)

Shahriari, B.; Swersky, K.; Wang, Z.; Adams, R. P.; de Freitas, N. *Proc. IEEE* **2016**, 104 (1), 148–175. (review)

Srinivas, N.; Krause, A.; Kakade, S. M.; Seeger, M. W. *IEEE Trans. Inf. Theory* **2012**, 58 (5), 3250–3265. (GP-UCB)

Rasmussen, C. E.; Williams, C. K. I. *Gaussian Processes for Machine Learning*; MIT Press: Cambridge, MA, 2006. (chapters 2 and 5)

## CHEMISTRY AND BENCHMARKS

Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Martinez Alvarado, J. I.; Janey, J. M.; Adams, R. P.; Doyle, A. G. *Nature* **2021**, 590 (7844), 89–96. (EDBO; the human benchmark)

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# References — categorical, multi-objective, noise

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